

ECHS CLAIM ANALYTICS

Discharge Delay & Audit Analysis

USE CASE 18 REPORT | FY 2025–2026 | 6.4M Claims

CLASSIFICATION	PERIOD	RECORDS ANALYSED	QUERY SETS
RESTRICTED	FY 2025–26	6.4M Claims	Use Case 18

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Summary:

Dataset scope: 6.4M FY25-26 claims. **83.86%** settled (**Rs 10,763 Cr**). **16.14%** pending (**Rs 4,456 Cr** exposure). Delay analysis on approved/e2e restricted to settled claims.

Discharge->Submission delay: Rejected claims submit 3 days later (median 9d vs 6d, $|r| \sim 0.38-0.41$, medium effect). Consistent with delayed billing at flagged hospitals.

Approval speed: Not-audited claims take 80 days median to settle - **4x** longer than audited-not-rejected (19d). Rejected claims: 62d. Audited-not-rejected on a separate fast administrative track.

UTI deductions vs delay: Effectively independent ($|r| < 0.07$). However, fast-approved claims ($\leq 15d$) are deducted only **0.36%** of the time vs **2.71%** for 121-180d claims - speed shields from scrutiny.

Deduction rates very low: Only **1.49%** of settled claims have any UTI deduction. Mean deduction = **0.099%** of billed amount. Audit-rejected claims have **13.62%** deduction rate (**9x** average).

Suspicious claims: 2,363 high-value fast-approved zero-deduction claims. **89%** (2,103) never audited. Top hospital: MAX Healthcare Dehradun (223 flagged, mean bill **Rs 2.1L**, 7-day approval).

Hospital patterns: Four risk tiers identified: (1) Systematic over-billers - ASG eye hospital chain; (2) Untouched high-billers - Vijay Hospital (6K claims, **Rs 2.3L** mean, zero scrutiny); (3) Double-flagged - JMC Medicity, Jain Medical, SDS Global; (4) Convergent fraud flag - Metro Hospital Haridwar (all four criteria). Dataset Overview

1. Dataset Overview

	Claims	Share	Mean Bill (Rs)	Exposure (Rs Cr)
Total FY25-26	64,04,135	100%	23,766	15,220
Settled	53,70,477	83.86%	20,042	10,763
Pending (unsettled)	10,33,658	16.14%	43,138	4,456
Audit-rejected	756	0.012%	1,69,385	13
Audited-not-rejected	3,31,345	5.17%	10,154-27,717	339
Not audited	60,72,034	94.8%	20,667-43,138	14868

Audit groups are defined as: (1) Audit-rejected - claims with a recovery amount raised in audit_query or audit_status (AQ_RECOVER_AMT > 0 or AS_RECOVER_AMT > 0); (2) Audited-not-rejected - claims appearing in audit tables but with zero recovery amount; (3) Not audited - no entry in either audit table.

2. Delay Analysis by Audit Category

With 6.4 million claims, any difference - however small - will produce $p \sim 0$. We therefore report median delays and rank-biserial effect sizes ($|r|$) as the meaningful measures. $|r| < 0.10$ = trivial · $0.10-0.30$ = small · $0.30-0.50$ = medium · > 0.50 = large

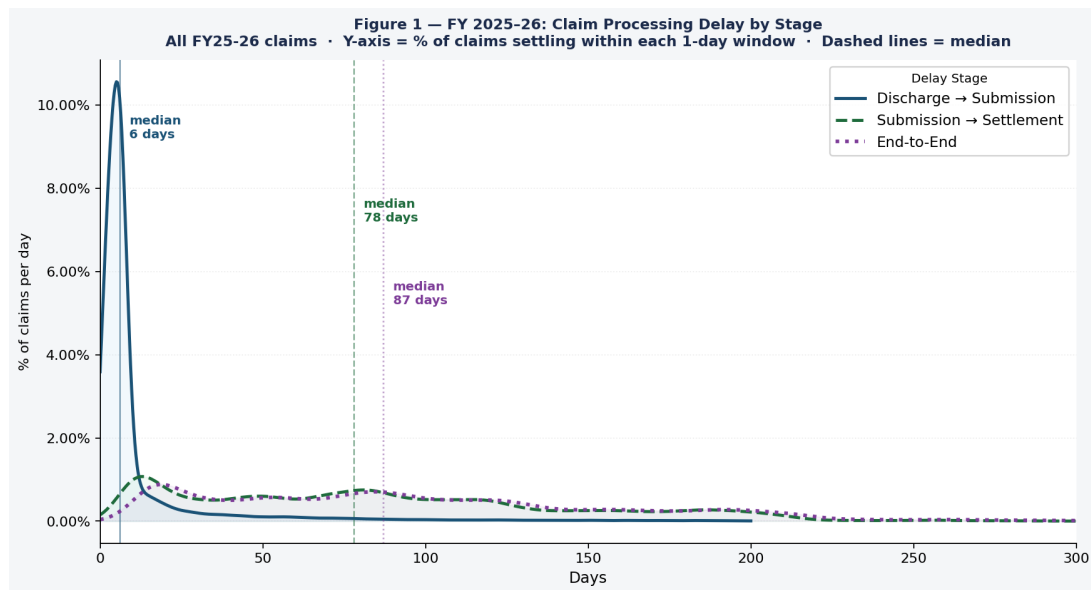
2a. Median Delays

Delay Segment	Audit-rejected	Audited-not-rejected	Not audited
Discharge -> Submission (all claims)	9 days	6 days	6 days
Submission -> Settlement (settled only)	62 days	19 days	80 days
End-to-End: Discharge -> Settlement	84 days	33 days	89 days

2b. Effect Sizes - Rank-biserial r (|r|)

- Discharge->Submission delay: Rejected claims take 3 extra days to submit (median 9 vs 6). Medium effect ($|r| \sim 0.38-0.41$) - consistent with delayed or padded billing at rejected hospitals.
- Submission->Settlement: Audited-not-rejected settles fastest (median 19 days) - these claims are processed through an administrative fast-track, distinct from fraud audit. Not-audited takes 80 days (standard queue). Rejected takes 62 days.
- End-to-End: Not-audited (89d) ~ Rejected (84d) >> Audited-not-rejected (33d). Trivial effect for Rejected vs Not-audited ($|r| = 0.09$), confirming the end-to-end difference is driven by the fast administrative track, not by fraud scrutiny.

2c. Distribution Plot - All Three Groups

**Figure 1 -- FY 2025-26: Claim Processing Delay by Stage**

Each curve shows what % of claims were resolved within a given number of days, with a taller, narrower peak meaning faster and more consistent processing.

3. Delay vs UTI Deduction Correlation

UTI deduction = gross_billed - net_paid (UTI's initial payment reduction). This is NOT the same as audit recovery (a separate process). Only 1.49% of settled claims (79,772) have any deduction.

Spearman correlations between delay and deduction percentage:

Delay and UTI deductions are effectively independent ($|r| < 0.10$ across all segments). However, the deduction rate by approval speed reveals a directional pattern:

Approval Window	Claims	% with Deduction	Mean Deduction %
0-15 days (fast track)	6,68,326	0.36%	0.032%
16-30 days	5,30,999	1.25%	0.073%
31-90 days	19,77,821	1.16%	0.089%
91-120 days	8,47,932	1.61%	0.126%
121-180 days	8,64,530	2.71%	0.152%
>180 days	4,80,853	2.25%	0.122%

- Fast-approved claims (0-15 days) are deducted only 0.36% of the time vs 2.71% for 121-180 day claims. Fast-tracking effectively shields claims from UTI scrutiny.
- This pattern is the basis of the suspicious claims flag - high-value claims approved in ≤ 10 days with zero deduction represent a statistically anomalous combination.

4. Deduction Analysis by Audit Group · settled claims only

Group	Settled Claims	% with Any Deduction	Mean Deduction %
Audit-rejected	749	13.62%	0.248%
Audited-not-rejected	3,29,653	0.48%	0.047%
Not audited	50,40,075	1.55%	0.103%
All settled	53,70,477	1.49%	0.099%

- Overall deduction rate is very low (1.49%). UTI pays nearly the full billed amount in 98.5% of cases.
- Rejected claims have 13.62% deduction rate - 9x higher than average - confirming that audit rejection and UTI cuts co-occur, though they are separate processes.
- Audited-not-rejected claims have the lowest deduction rate (0.48%), consistent with administrative fast-tracking rather than rigorous scrutiny.

5. Suspicious Claims Flag · fast approval + high bill + zero deduction

Threshold: approval delay ≤ 10 days (p5 of all settled claims) AND gross billed $\geq$ Rs 1,20,202 (p95 of all settled claims) AND zero UTI deduction. Rationale: Flags claims in the fastest 5% of settlements AND top 5% by bill value that received no reduction. High-value claims approved in under 2 weeks without scrutiny are a statistically anomalous intersection.

Metric	Value
Approval delay p5 (threshold)	10 days
Gross billed p95 (threshold)	Rs 1,20,202
Suspicious claims flagged	2,363 (0.044% of settled)
Of which: also audit-rejected	15
Of which: never audited	2,103 (89%)

Top hospitals by suspicious claim count:

Hospital	Flagged	Mean Bill (Rs)	Mean Approval (days)	Also Rejected
MAX HEALTH CARE INSTITUTE - DEHRADUN	223	2,12,493	7	2
HOMI BHABHA CANCER HOSPITAL - MULLANPUR	167	3,65,672	6	0
SWAMI RAMA HIMALAYAN UNIVERSITY	156	2,08,740	8	0
SHRI MAHANT INDIRESH HOSPITAL	133	2,15,417	8	3
AMRITA INSTITUTE OF MEDICAL SCIENCES	122	3,30,981	8	0
SHIVALIK MEDICAL CENTRE - NOIDA	102	1,40,995	6	0
GRAPHIC ERA INSTITUTE OF MEDICAL SCIENCES	99	2,29,442	8	0
HBCH&MPMMCC - TATA MEMORIAL VARANASI	74	4,80,436	7	0

- 89% of suspicious claims (2,103) were never audited - high-value fast-approvals that bypassed scrutiny entirely.
- Cancer and super-speciality hospitals dominate this list, suggesting high-cost specialty procedures are processed on fast administrative tracks.

6. Hospital Risk Patterns · four categories

6a. Systematic Over-billing: Highest Mean UTI Deduction %

These hospitals face consistent UTI payment cuts across all their claims - pattern suggests systematic over-billing.

Hospital	Claims	Mean Deduction %	% Claims Deducted	Rejection Rate
ASG HOSPITAL - PIMPLE SOUDAGAR	272	12.3%	21.7%	0%
ASG HOSPITAL - PUNE	973	11.1%	19.2%	0%
AJAY HOSPITAL - MEERUT	616	9.9%	99.2%	0.16%
NEW DELHI CENTRE FOR SIGHT - KUKATPALLY	185	7.3%	13.0%	0%
SARAN HOSPITAL	374	7.0%	54.8%	0%
PURUSHOTTAMDAS CANCER CARE CENTRE	602	6.0%	100.0%	0%
BUDDHA CANCER CENTRE	412	6.0%	74.0%	0%

- ASG Hospital chain (eye hospitals) appears in multiple cities with 6-12% mean deduction. UTI is systematically cutting their bills, yet rejection rate is 0% - audit not being triggered.

6b. Untouched High-billers: High Bill, Zero Deduction, Never Audited

High-value hospitals where UTI has never deducted a rupee and no audit action has been taken.

Hospital	Claims	Mean Bill (Rs)	Deduction %	Rejection Rate
VIJAY HOSPITAL	6,178	2,30,256	0%	0%
VIAN MULTISPECIALITY HOSPITAL	121	1,90,691	0%	0%
VINAYAK MEDICARE HOSPITAL	678	1,91,614	0%	0%
VASAL HOSPITAL PVT LTD	377	1,65,648	0%	0%
RPS HOSPITAL MATERNITY & TRAUMA	351	1,67,323	0%	0%
RLKC HOSPITAL METRO HEART - NEW DELHI	166	3,48,744	0%	0%
UMANG SUPERSPECIALITY HOSPITAL	595	1,52,148	0%	0%
VEDANTA HOSPITAL AND INSTITUTE	1,751	1,16,559	0%	0%

- RLKC Hospital New Delhi: mean bill Rs 3.49L, 166 claims, zero deduction, zero rejection.
- Vijay Hospital: 6,178 claims, mean bill Rs 2.3L - largest untouched high-biller by volume.

6c. Double Red Flag: High Rejection Rate AND High Deduction

Hospital	Claims	Rejection Rate	Mean Deduction %	Med Appr Delay
JMC MEDICITY	1,756	0.57%	0.57% (100% claims deducted)	106d
JAIN MEDICAL CENTER & LASER EYE	2,835	0.53%	0.82% (97.6% deducted)	110d

Hospital	Claims	Rejection Rate	Mean Deduction %	Med Appr Delay
SDS GLOBAL HOSPITAL	2,112	0.38%	3.55% (99.7% deducted)	112d
SANTOSH HOSPITAL - MEERUT	1,043	0.29%	2.00% (100% deducted)	121d
VALENTIS CANCER CARE HOSPITAL	1,043	0.29%	1.28% (99.4% deducted)	114d
MEDICOVER HOSPITALS - VENKOJIPALEM	2,399	0.33%	0.54%	110d

6d. Convergent Fraud Pattern: Fast Approval + High Bill + No Deduction + High Rejection

METRO HOSPITAL & HEART INSTITUTE - HARIDWAR

6e. Rejection Rate Ranking (min 20 audited claims)

Hospital	Total Claims	Rejected	Rej Rate	Med Sub Delay	Med Appr Delay
METRO HOSPITAL - HARIDWAR	578	15	2.60%	5d	14d
RIDDHI VINAYAK CARDIAC CENTRE	174	4	2.30%	53d	133d
APEX HOSPITAL - MULUND	690	8	1.16%	7d	90d
KAIZEN SUPER SPECIALITY HOSPITAL	100	1	1.00%	20d	149d
KANISHK SURGICAL & SUPERSPECIALITY	5,966	43	0.72%	8d	13d
VELMED HOSPITAL	11,298	55	0.49%	8d	14d
JAIN MEDICAL CENTER & LASER EYE	3,117	15	0.48%	7d	110d

Appendix - Statistical Tests & Effect Sizes

Use Case 18 · ECHS FY 2025-26 · Full dataset: n = 64,04,135 claims ·

Step 1 - Why Non-Parametric Tests?

The skew evidence below confirms the non-parametric choice (Kruskal-Wallis + Mann-Whitney) is correct.

Delay Segment	Group	n	Median	Mean	Skew Evidence (Mean >> Median)
Discharge->Submission	Audit-Rejected	756	9d	32.7d	YES - 3.6x

Delay Segment	Group	n	Median	Mean	Skew Evidence (Mean >> Median)
Discharge->Submission	Audited-Not-Rejected	3,31,345	6d	14.3d	YES - 2.4x
Discharge->Submission	Not Audited	60,72,034	6d	13.7d	YES - 2.3x
Submission->Settlement	Audit-Rejected	749	62d	58.8d	Mild
Submission->Settlement	Audited-Not-Rejected	3,29,653	19d	43.8d	YES - 2.3x
Submission->Settlement	Not Audited	50,40,075	80d	88.0d	Mild
End-to-End	Audit-Rejected	749	84d	91.6d	Mild
End-to-End	Audited-Not-Rejected	3,29,653	33d	58.1d	YES - 1.8x
End-to-End	Not Audited	50,40,075	89d	100.3d	Mild

Step 2 - Kruskal-Wallis Omnibus Test (3 groups)

Delay Segment	H statistic	p-value	Significant?
Discharge -> Submission	1,131.2	2.32 x 10-246	YES
Submission -> Settlement	2,12,454.2	< 10-300	YES
End-to-End	1,70,578.1	< 10-300	YES

Interpretation: All three delays show significant group differences. However, $H = 2,12,454$ for approval delay does NOT mean the effect is large - it means the sample is enormous. The H statistic scales with n . Effect size ($|r|$) is the correct measure of practical importance.

Step 3 - Pairwise Mann-Whitney U Tests with Benjamini-Hochberg FDR Correction

Compares each pair of groups separately. 9 comparisons in total (3 pairs x 3 delay types). Running 9 tests inflates false-positive risk - Benjamini-Hochberg (BH) correction adjusts p-values to control the False Discovery Rate at 5%. BH was chosen over Bonferroni because it is less conservative and more powerful for correlated tests. The reported p_{BH} values are already corrected.

Delay	Group 1	Group 2	n1	n2	BH-adjusted p	r	Magnitude
Sub delay	Rej	Aud-not-rej	756	3,31,345	1.21×10^{-71}	0.376	Medium
Sub delay	Rej	Not audited	756	60,72,034	5.25×10^{-85}	0.409	Medium
Sub delay	Aud-not-rej	Not audited	3,31,345	60,72,034	6.87×10^{-165}	0.028	Trivial
Appr delay	Rej	Aud-not-rej	749	3,29,653	3.13×10^{-38}	0.273	Small
Appr delay	Rej	Not audited	749	50,40,075	2.68×10^{-39}	0.277	Small
Appr delay	Aud-not-rej	Not audited	3,29,653	50,40,075	~ 0	0.478	Medium
E2E delay	Rej	Aud-not-rej	749	3,29,653	1.81×10^{-69}	0.372	Medium
E2E delay	Rej	Not audited	749	50,40,075	3.02×10^{-5}	0.088	Trivial
E2E delay	Aud-not-rej	Not audited	3,29,653	50,40,075	~ 0	0.429	Medium

All 9 comparisons are significant after BH correction. This is expected - not a finding. With $n1 \times n2$ in the billions for some pairs, the Mann-Whitney U test has effectively infinite power. The |r| column is the actual result of interest.

Step 4 - Effect Size: Rank-Biserial r

Rank-biserial $r = 1 - 2U / (n1 \times n2)$, derived from the Mann-Whitney U statistic. Equivalent to: $P(\text{Group 1 delay} > \text{Group 2 delay}) - P(\text{Group 1 delay} < \text{Group 2 delay})$. Independent of sample size - does not inflate with larger n. Thresholds follow Cohen (1988): $|r| < 0.10$ trivial · $0.10-0.30$ small · $0.30-0.50$ medium · > 0.50 large.

Delay	Comparison	r	Magnitude	Direction (longer delay)
Discharge->Submission	Rej vs Aud-not-rej	0.376	Medium	Audit-Rejected
Discharge->Submission	Rej vs Not-audited	0.409	Medium	Audit-Rejected
Discharge->Submission	Aud-not-rej vs Not-aud	0.028	Trivial	Negligible
Submission->Settlement	Rej vs Aud-not-rej	0.273	Small	Audit-Rejected
Submission->Settlement	Rej vs Not-audited	0.277	Small	Not Audited
Submission->Settlement	Aud-not-rej vs Not-aud	0.478	Medium	Not Audited
End-to-End	Rej vs Aud-not-rej	0.372	Medium	Audit-Rejected
End-to-End	Rej vs Not-audited	0.088	Trivial	Negligible
End-to-End	Aud-not-rej vs Not-aud	0.429	Medium	Not Audited

Step 5 - Spearman Correlation: Delay vs UTI Deduction %

Tests monotonic relationship between delay (days) and deduction % $((\text{gross_billed} - \text{net_paid}) / \text{gross_billed} \times 100)$.

Delay Segment	Sample	n	Spearman ρ	p-value
Discharge->Submission	Deducted claims only	79,772	-0.097	3.51×10^{-164}
Submission->Settlement	Deducted claims only	79,772	-0.024	1.46×10^{-11}
End-to-End	Deducted claims only	79,772	-0.041	3.44×10^{-31}
Discharge->Submission	All settled	53,70,477	+0.018	~ 0
Submission->Settlement	All settled	53,70,477	+0.051	~ 0
End-to-End	All settled	53,70,477	+0.061	~ 0

All $|\rho| < 0.10$ across every combination. Delay and UTI deductions are effectively independent. The p-values are again significant only because of sample size - a ρ of 0.018 with $n = 5.37M$ will always be "significant" but explains 0.03% of variance. Conclusion: no meaningful linear or monotonic relationship exists between delay and deduction.